PCY Algorithm

PCY (Park-Chen-Yu) Algorithm

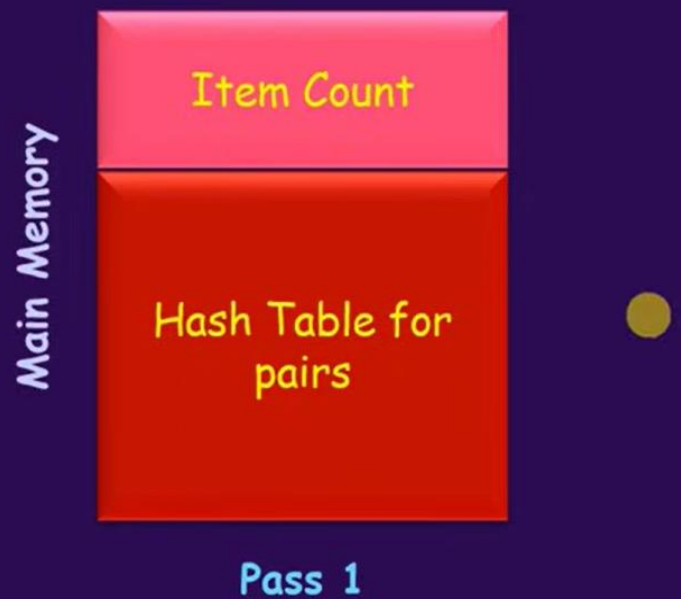
Overview

- Developed to efficiently find frequent itemsets in large datasets
- Algorithms like Apriori were computationally expensive for with large datasets.
- This was due to need to generate and test a large number of candidate itemsets.
- Uses a hash-based approach to count itemset frequency and prune infrequent itemsets.
- This avoids the need to generate candidate itemsets, making the algorithm more efficient.
- It is commonly used in data mining and association rule learning.
- Applications of the PCY algorithm include market basket analysis, recommendation systems, and web log analysis.

PCY (Park-Chen-Yu) Algorithm

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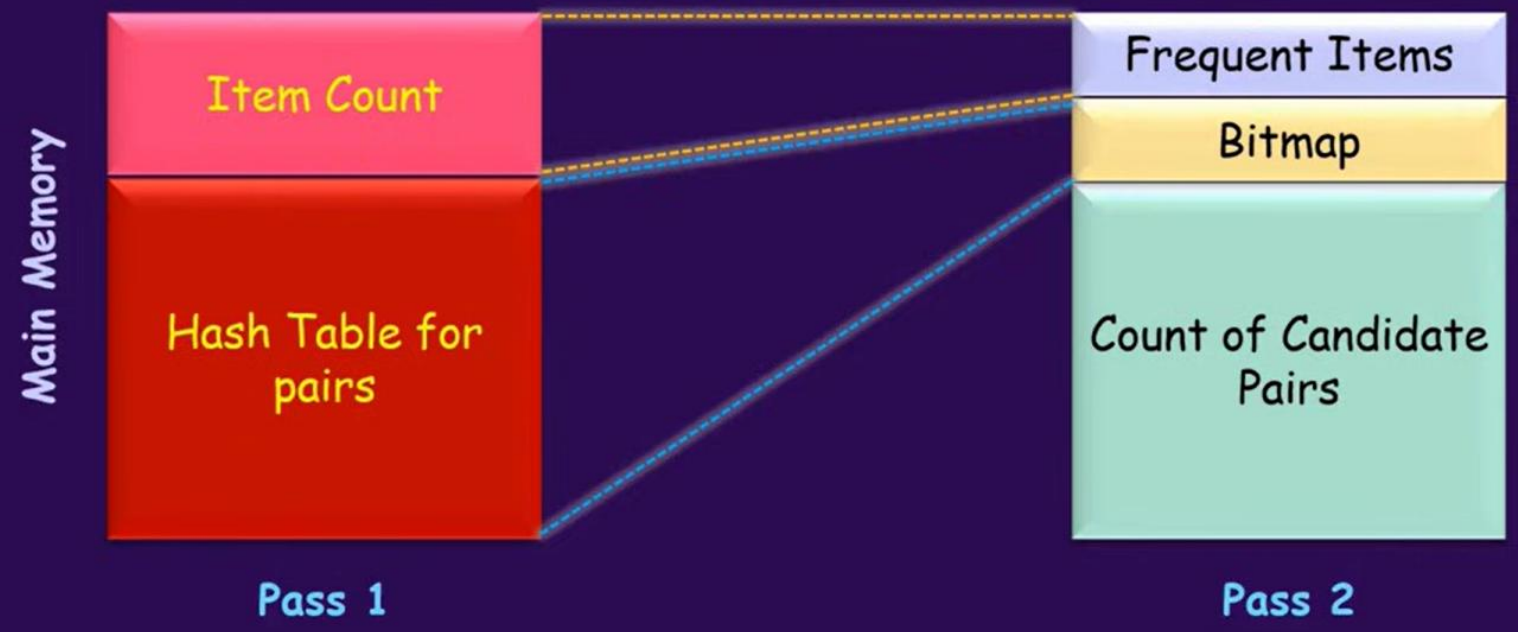
Main Memory: Picture of PCY



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PCY (Park-Chen-Yu) Algorithm

Main Memory: Picture of PCY



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PCY (Park-Chen-Yu) Algorithm

Algorithm

```
Pass 1:
    FOR (each basket):
        FOR (each item in the basket):
            add 1 to item's count
        FOR (each pair of items):
            hash the pair to a bucket
            add 1 to the count for that bucket
```

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Pass 2:

Count all pairs {i, j} that meet the conditions for being a candidate pair:

1. Both i and j are frequent items
2. The pair {i, j} hashes to a bucket whose bit in the bit vector is 1

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Example

Find frequent item pairs using PCY algorithm considering threshold as 2

Transactions	Items
T1	1, 2, 3
T2	4, 5
T3	1, 4, 5
T4	1, 2, 4
T5	3, 4, 5
T6	2, 4, 5

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Example

Step 1: Find length of each item and eliminate items whose count is less than 1

Transactions	Items
T1	1, 2, 3
T2	4, 5
T3	1, 4, 5
T4	1, 2, 4
T5	3, 4, 5
T6	2, 4, 5

1	3
2	3
3	2
4	5
5	4

Eliminate those items whose count is less than 1, but in our case no item count is less than 1.

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Example

Step 1: Find length of each item and eliminate items whose count is less than 1

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T1	1, 2, 3
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1	3
2	3
3	2
4	5
5	4

Eliminate those items whose count is less than 1, but in our case no item count is less than 1.

Candidate Item-set:
{1, 2, 3, 4, 5}

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Example

Step 2: Form key-value pair where key is the item-pair and value is their occurrence

Transactions	Items	Pairs with their counts
T1	1, 2, 3	{(1,2) : 2}, {(2,3) : 1}, {(1,3) : 1}
T2	4, 5	
T3	1, 4, 5	
T4	1, 2, 4	
T5	3, 4, 5	
T6	2, 4, 5	

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Example

Step 2: Form key-value pair where key is the item-pair and value is their occurrence

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T3	1, 4, 5	{(1,4) : 2}, {(1,5) : 1}
T4	1, 2, 4	{(2,4) : 2}
T5	3, 4, 5	{(3,4) : 1}, {(3,5) : 1}
T6	2, 4, 5	{(2,5) : 1}

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Example

Step 3: Eliminate pairs whose count of occurrence does not satisfy threshold value

Transactions	Pairs
T1	{(1,2) : 2}, {(2,3) : 1}, {(1,3) : 1}
T2	{(4,5) : 4}
T3	{(1,4) : 2}, {(1,5) : 1}
T4	{(2,4) : 2}
T5	{(3,4) : 1}, {(3,5) : 1}
T6	{(2,5) : 1}

Since, threshold is 2, hence eliminate those pairs whose count is less than 2

Final Pairs:
{(1,2), (4,5), (1,4), (2,4)}

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Example

Step 4: Find bucket numbers

Final Pairs: {(1,2), (4,5), (1,4), (2,4)}

Now, Apply hash function to find bucket no. :

For pair (i,j), hash function will be **(i * j) mod 10**

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Example

Step 4: Find bucket numbers

Final Pairs: {(1,2), (4,5), (1,4), (2,4)}

Now, Apply hash function to find bucket no. :

For pair (i,j), hash function will be **(i * j) mod 10**

Pairs	Bucket No.
(1,2)	(1 * 2) mod 10 = 2
(4,5)	(4 * 5) mod 10 = 0
(1,4)	(1 * 4) mod 10 = 4
(2,4)	(2 * 4) mod 10 = 8

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Example

Step 5: Create Candidate set table

Bit Vector	Bucket No	Count	Pairs	Candidate Set

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Example

Step 5: Create Candidate set table

Bit Vector	Bucket No	Count	Pairs	Candidate Set
	2	2	(1,2)	
	0	4	(4,5)	
	4	2	(1,4)	
	8	2	(2,4)	

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Example

Step 5: Create Candidate set table

Bit Vector	Bucket No	Count	Pairs	Candidate Set
1	2	2	(1,2)	(1,2)
1	0	4	(4,5)	(4,5)
1	4	2	(1,4)	(1,4)
1	8	2	(2,4)	(2,4)

Therefore, Candidate pair set: {(1,2), (4,5), (1,4), (2,4)}

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Multistage PCY - Refinement of PCY algorithm

Why use Multistage PCY?

1. More Hash Tables: Multistage uses several successive hash tables to further reduce the number of candidate pairs.

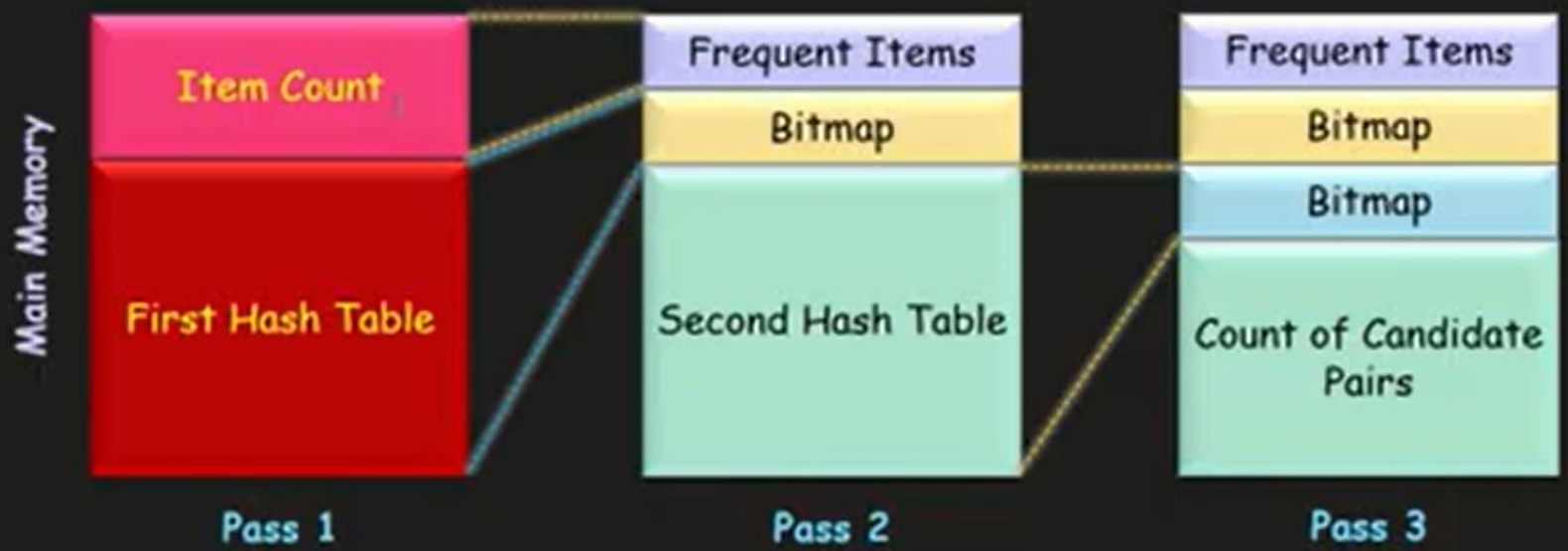
2. Additional Passes: Unlike PCY, which takes two passes to find frequent pairs, Multistage takes more than two passes. This allows it to refine the candidate pairs more effectively.

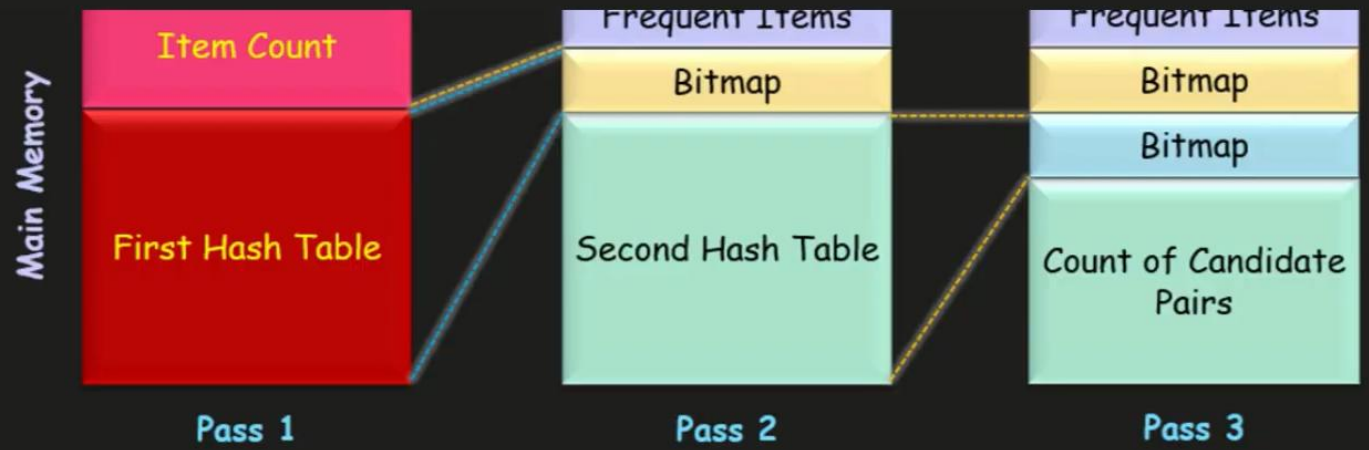
3. Memory Utilization: In Multistage, bit-vectors eventually consume all of the main memory. This is a trade-off for the increased accuracy in candidate pair selection.

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Algorithm

1. Pass 1: Count items and Hash pairs {i, j}
2. Pass 2:
   if (i, j are frequent) and ({i, j} hashes to frequent bucket in B1):
   Hash pairs {i, j} into Hash2
3. Pass 3:
   if (i,j are frequent) and ({i,j} hashes to freq. bucket in B1) and ({i,j} hashes to freq. bucket in B2):
   Count pairs {i,j}

Example

Basket 1: {apple, banana, mango} -> {A, B, M}
Basket 2: {apple, orange} -> {A, O}
Basket 3: {banana, mango} -> {B, M}
Basket 4: {apple, banana} -> {A, B}
Basket 5: {mango, orange} -> {M, O}

Minimum support count to be 2.

Example

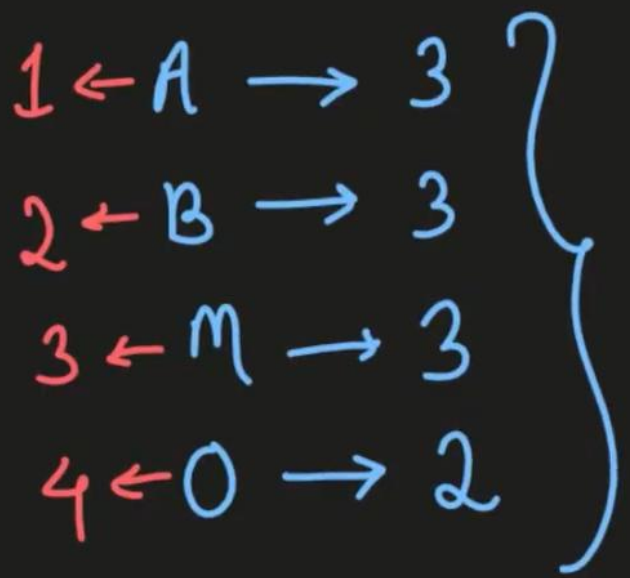
Basket 1: {apple, banana, mango} -> {A, B, M}
Basket 2: {apple, orange} -> {A, O}
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Basket 4: {apple, banana} -> {A, B}
Basket 5: {mango, orange} -> {M, O}
Minimum support count to be 2.

A → 3
B → 3
M → 3
O → 2

Example

Basket 1: {apple, banana, mango} -> {A, B, M}
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Minimum support count to be 2.

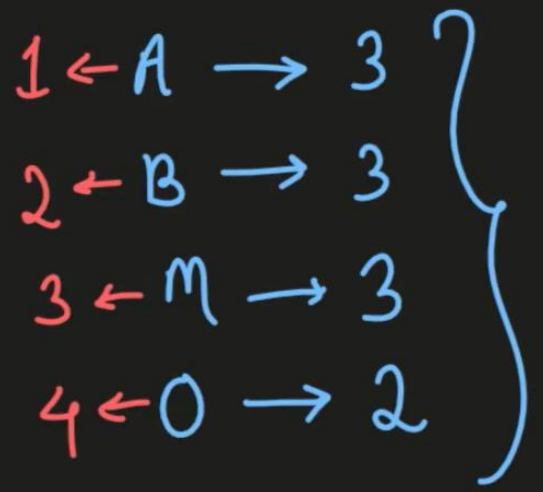


Example

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Basket 1: {apple, banana, mango} -> {A, B, M}
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Minimum support count to be 2.



$$\text{Hash } 1 = (i + j) \bmod 3$$

$$\text{Hash 1} = (i + j) \bmod 3$$

$(A, B) \rightarrow (1, 2) \rightarrow (1 + 2) \bmod 3 = 0$

$(B, M) \rightarrow (2, 3) \rightarrow (2 + 3) \bmod 3 = 2$

$(A, M) \rightarrow (1, 3) \rightarrow (1 + 3) \bmod 3 = 1$

$(A, 0) \rightarrow (1, 4) \rightarrow (1 + 4) \bmod 3 = 2$

$(M, 0) \rightarrow (3, 4) \rightarrow (3 + 4) \bmod 3 = 1$

(A, B) → (1, 2) → (1 + 2) mod 3 = 0

(B, M) → (2, 3) → (2 + 3) mod 3 = 2

(A, M) → (1, 3) → (1 + 3) mod 3 = 1

(A, 0) → (1, 4) → (1 + 4) mod 3 = 2

(M, 0) → (3, 4) → (3 + 4) mod 3 = 1

Bucket 0 → 1 pair

Bucket 1 → 2 pairs

Bucket 2 → 2 pairs

Bucket 0 → 1 pair

Bucket 1 → 2 pairs

Bucket 2 → 2 pairs

Frequent Buckets

1, 2

Bucket 2 → 2 pairs | 1, 2

Pass 2 :

Hash 2 = (i * j) mod 3

(B, m) → (2, 3) = (2 x 3) mod 3 = 0

(A, m) → (1, 3) = (1 x 3) mod 3 = 0

(A, 0) → (1, 4) = (1 x 4) mod 3 = 1

(M, 0) → (3, 4) = (3 x 4) mod 3 = 0

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$(A,10) \rightarrow (1,4) = (1 \times 4) \bmod 3 = 1$

$(M,10) \rightarrow (3,4) = (3 \times 4) \bmod 3 = 0$

Bucket 0 $\rightarrow$ (A,M), (B,M) and (M,10) $\rightarrow$ 3 pairs

Bucket 1 $\rightarrow$ (A,10) $\rightarrow$ 1 pair X

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Bucket 0 → (A,m), (B,m) and (m,0) → 3 pairs ✓

Bucket 1 → (A,0) → 1 pair ✗

Pass 3 :-

(A,m)
(B,m)
(m,0)

} Selected pair / Candidate pairs

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